

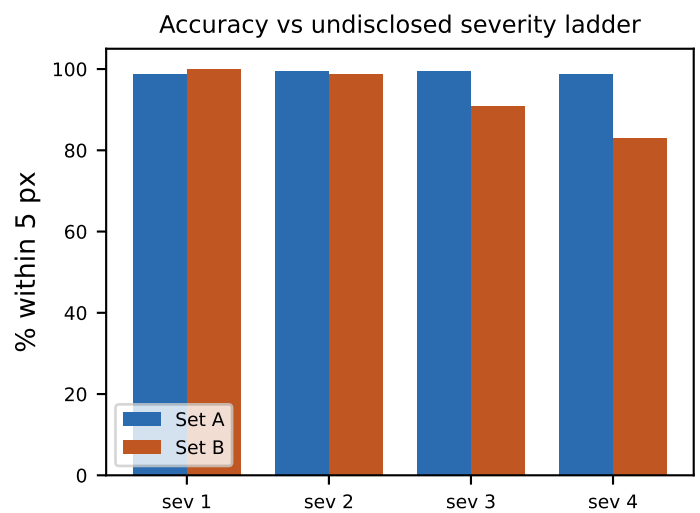
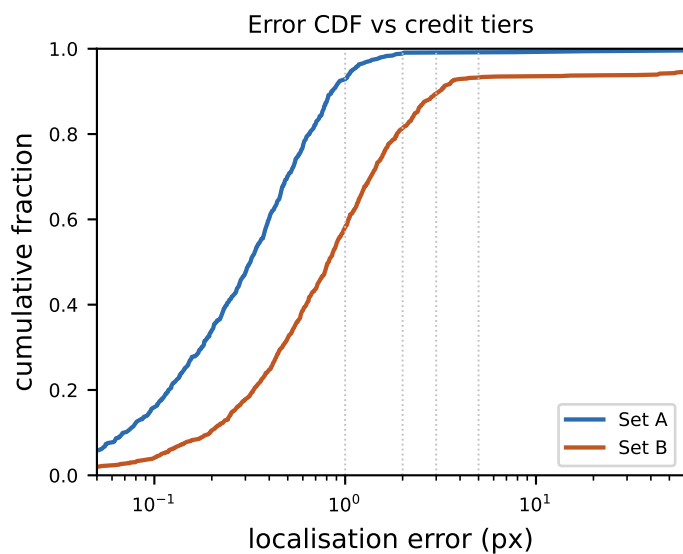
# Drift-Sense Phase 2 — Failure Analysis

Measured on 2250 pairs from an independently generated Phase 2 set (Set A 875, Set B 875, Set C 500 absent, Set D 0). This generator is not ours, so these numbers include a domain gap that a self-generated split cannot show.

## 1. Where localisation credit is lost

$\leq 1$  px (credit 1.00): 75.4%   1-2 px (0.80): 14.7%   2-3 px (0.60): 4.1%   3-5 px (0.40): 1.9%    $> 5$  px (0.00): 3.9%

Set A credit 0.978, Set B credit 0.830; weighted  $0.45 \cdot A + 0.55 \cdot B = 0.897$ . The dominant loss is not the  $>5$  px tail (3.9% of present pairs) but the traffic between the 1 px and 3 px tiers, which is a precision problem, not a search problem.



## 2. What the residual $>5$ px failures actually are

68 of 1750 present pairs (3.9%) land beyond 5 px, and 88% of them are Set B (Set B fails at 6.9% against Set A's 0.9%).

Standardised separation between failures and successes, largest first:

drift jitter  $d=+1.46$ , speckle  $d=+1.46$ , detector noise  $d=+1.42$ , charging  $d=+1.42$ , shear  $d=+1.41$

These are dominated by acquisition severity, not by pose. The distinction is worth stating because the previous round's localisation failures were all wrong scale-basin lock-ons, so the natural assumption was that these are the same thing. They are not: every acquisition term separates the two groups at  $d \approx 1.1$ , while  $|\text{rotation}|$  manages only  $d=+0.20$  and polygon scaling essentially zero. Rotation is therefore a secondary effect at most, and polygon scaling — the one Set B degradation we had not modelled — turns out not to be what Set B is failing on. Both were candidate fixes before the separation was measured; neither is where the points are.

## 3. Rejection and calibration

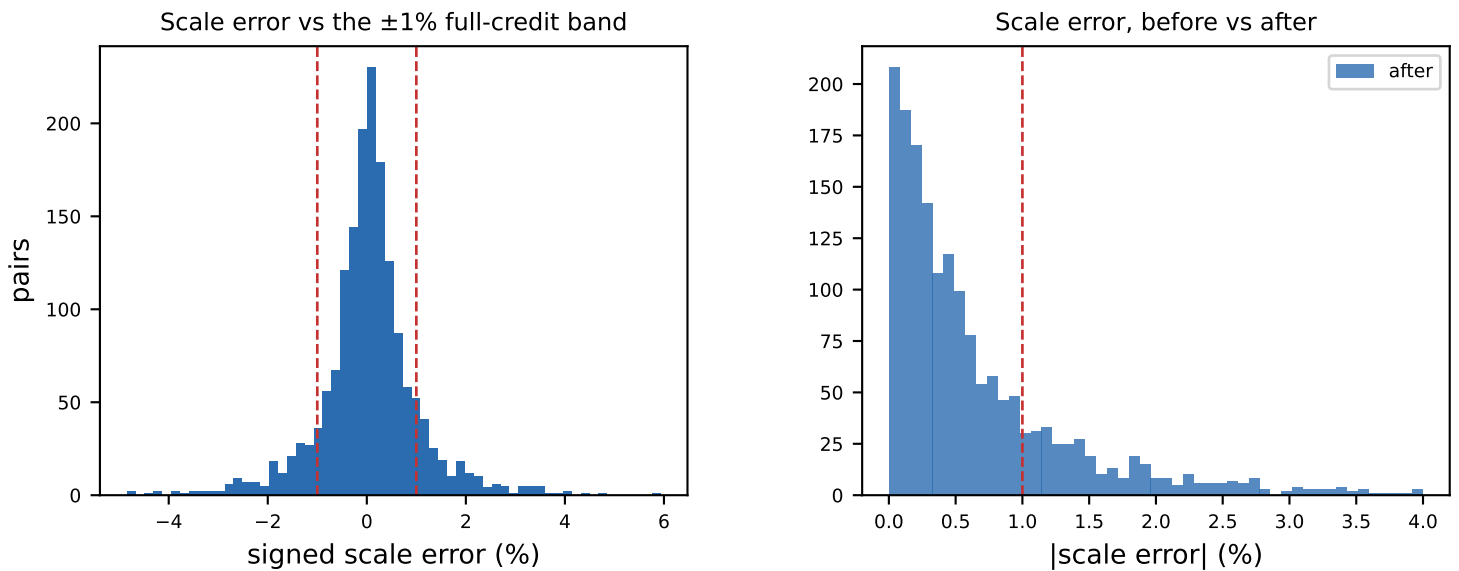
At the shipped threshold 0.18: 459 correct rejections, 39 real instances wrongly rejected, 41 absent pairs wrongly accepted. F1 (reject as the positive class) = 0.920.

The positive class matters. An always-found system scores F1 = 0.875 if 'present' is positive but exactly 0.000 if 'reject' is positive. The rubric states that never rejecting scores zero, so the reject-positive convention is the binding one and is what is reported here.

## 4. The scale-quantisation defect, and its fix

The template was rendered with `cv2.resize` to `round(reference_px / m)`, so the magnification the template actually realised was `reference_px / round(reference_px / m)` — only 43 attainable values across `[8, 12]`, in steps 0.81-1.22% wide. The Phase 2 scale tier pays full credit below 1%, so the quantisation step was as wide as the entire full-credit band, and any search over `m` was optimising a piecewise-constant function. This is why an earlier attempt to polish scale made it worse and had to be disabled: golden-section search on a staircase returns an arbitrary point on a plateau.

Fix: apply the residual sub-integer scale as part of the affine that was already being paid for to apply rotation. Measured realisation error fell from a median of 0.26% (worst 0.55%) to 0.012% (worst 0.021%), at no extra resampling cost, and the nominal 10× path stays bit-identical. A second bias was removed alongside it: `TM_CCOEFF_NORMED` is normalised over the template's own support, so candidates of different pixel counts are not comparable and the search was pulled towards larger magnification. The polish stage now pins the template canvas across its sweep.



## 5. What we could not fix, and why

The 1 px tier is bounded by the label, not by the method. Per-row drift jitter in the generator has  $\sigma \approx 0.94\text{--}0.99$  px and is unlearnable: the error signature matches it exactly, with `dx` scattering  $\sim 1.0$  px while `dy` scatters  $\sim 0.10$  px, because raster jitter displaces rows horizontally. A rigid template cannot align to a row-by-row distorted frame, so a  $\sim 68\%$  share within 1 px is the 1-sigma outcome. Undoing the row warp before matching is the only route past it, and only the  $\sim 83\text{--}125$  rows under the template carry usable signal, so the payoff is uncertain. Whether the graders' generator carries the same drift model is unknown.

Additional pose hypotheses are exhausted: `K=5` returns results identical to `K=3`, because the coarse scale sweep produces only about three local maxima.

Honest limits of this document: the blind set is 200 pairs, this measurement is a few hundred, and both the noise ladder and the severity parameters are undisclosed. Set D (optical, RGB) is scored here but was never trained for.